

07 — ROI Analysis

Health Hub Business Plan v2 | Return on Investment Analysis Version: 2.0 | Date: March 2026

All assumptions, cost figures, revenue projections, and scenario definitions in this document trace back to [params.md](#). Revenue trajectories reference [05-revenue-modelling](#) and cost data references [03-cost-breakdown](#). Where a number appears without inline citation, it is derived from those sources.

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1. Executive Summary

This document presents a comprehensive return-on-investment analysis for Health Hub across all 21 scenario combinations defined in params.md. The analysis is designed for a pre-revenue health technology startup targeting East Africa (Ethiopia first, Kenya second) and therefore differs from traditional ROI frameworks in several important ways:

- **No historical revenue baseline exists.** All revenue figures are forward projections based on bottom-up user modelling and top-down market sizing.
- **Capital investment is a mix of economic cost and cash outlay.** Founder labor is counted at market-equivalent rates (\$10-\$20/hr per params.md), not at zero.
- **The time horizon matters more than a single ROI number.** A 36-month view is required because revenue collection does not begin until Pilot 2 (Month 5-10 depending on path), and cumulative crossover occurs even later.

Key Findings

Finding	Detail
Highest capital-efficiency ROI at M24	S1-L3 (self-funded, fully resourced): 92% cumulative Rev/CapEx ratio
Fastest absolute payback (investor paths)	S3-L3-I1 and S3-L2-I1: success metric reached at M18-M22
Baseline plan (S2-O2-I2) ROI at M36	Projected 170-220% cumulative ROI for combined capital
Best risk-adjusted path	S3-O2-I2: 20-month timeline, \$215K total capital, strong growth curve
Paths to avoid	S2-O1-I3 and S3-O1-I3: large capital sits idle during slow L1 development
Investor IRR range (36-month)	15-85% depending on scenario, with S3-L3-I2 offering the best investor-specific return
Founder ROI advantage	Self-funded paths (S1) deliver highest percentage ROI but lowest absolute returns

Framework for Reading This Document

ROI is presented across three return horizons for each scenario family:

- **12 months post-production launch** — early traction signal
- **24 months post-production launch** — medium-term validation
- **36 months post-production launch** — strategic return horizon

For investor-involved scenarios (S2 and S3), founder ROI and investor ROI are tracked separately because their capital enters at different times, at different amounts, and their ownership stakes differ.

2. Methodology

2.1 ROI Formula

The primary ROI calculation used throughout this document:

ROI = (Cumulative Revenue – Cumulative Investment) / Cumulative Investment x 100%

Where:

- **Cumulative Revenue** = total platform revenue collected from all streams (GP consults, specialist consults, pharmacy commissions, diagnostics commissions, platform fees, subscriptions) from project start through the measurement date.
- **Cumulative Investment** = total capital deployed (founder capital + investor capital where applicable) through the measurement date.

2.2 Dual ROI Tracking

For scenarios involving outside investors (S2 and S3), two separate ROI calculations are maintained:

Founder ROI:

Founder ROI = (Founder Share of Cumulative Revenue – Founder Capital Invested) / Founder Capital Invested x 100%

Where "Founder Share of Cumulative Revenue" is the founder's pro-rata equity percentage (60% in S1, 55-60% in S2/S3 per params.md Section 8.3) multiplied by cumulative revenue. This is a simplification that assumes revenue accrues to equity holders proportionally — in practice, founder returns would be realized through salary, dividends, or exit events.

Investor ROI:

Investor ROI = (Investor Share of Cumulative Revenue – Investor Capital) / Investor Capital x 100%

Where "Investor Share of Cumulative Revenue" is the investor's equity percentage (10-20% drawn from the investor pool per params.md Section 8.2) multiplied by cumulative revenue.

2.3 Economic Cost vs Cash ROI

This document calculates ROI against **economic cost** (full market-value cost of all labor, infrastructure, and operations) rather than cash-only spend. This is the conservative approach because:

- It produces lower ROI numbers than cash-only analysis
- It reflects the true cost of building the business, including founder sweat equity
- It is the basis investors will use to evaluate the opportunity

Per params.md, the total economic cost ranges from \$230,522 (S3-O3-I3) to \$451,751 (S1-L1), while founder capital (cash deployed) ranges from \$95,000 to \$240,000.

2.4 Revenue Attribution

Revenue is attributed to the period in which it is collected. Per the revenue model (05-revenue-modelling):

- **Pre-Pilot and Pilot 1:** \$0 revenue (free or symbolic pricing)
- **Pilot 2:** Revenue at 50% of target pricing, beginning Month 5-10 depending on path
- **Production:** Revenue at full pricing, beginning Month 9-22 depending on path
- **Post-Production extension (Year 3+):** Projected based on 15% monthly growth rate (moderate assumption) with 7% monthly churn

2.5 Important Caveats

1. All revenue projections use the **moderate** assumption set unless otherwise stated. Conservative and optimistic bounds are provided in the sensitivity analysis (Section 8).
2. ROI calculations do not account for the time value of money. A discounted ROI analysis would reduce all figures by approximately 10-15% at a 10% discount rate over 36 months.
3. "Revenue" in this context means gross platform revenue before operating expenses. The platform's blended margin is 91-96% on B2C transactions (per 05-revenue-modelling), so net revenue closely tracks gross revenue.
4. Exit-based ROI (e.g., company sale or Series A valuation) is not modeled. The ROI figures here reflect organic revenue returns only.

3. Investment Totals by Scenario

3.1 Complete Investment Matrix — All 21 Combinations

#	Code	Founder Capital	Investor Capital	Total Capital	Economic Cost	Timeline (Months)	Cash vs Economic Gap
1	S1-L1	\$240,000	\$0	\$240,000	\$451,751	34	\$211,751
2	S1-L2	\$185,000	\$0	\$185,000	\$353,794	26	\$168,794
3	S1-L3	\$150,000	\$0	\$150,000	\$281,931	20	\$131,931
4	S2-O1-I1	\$120,000	\$140,000	\$260,000	\$400,716	30	\$140,716
5	S2-O1-I2	\$120,000	\$145,000	\$265,000	\$363,656	28	\$98,656
6	S2-O1-I3	\$120,000	\$245,000	\$365,000	\$326,066	24	-\$38,934
7	S2-O2-I1	\$110,000	\$135,000	\$245,000	\$382,548	28	\$137,548
8	S2-O2-I2	\$110,000	\$120,000	\$230,000	\$331,570	24	\$101,570
9	S2-O2-I3	\$110,000	\$240,000	\$350,000	\$307,898	22	-\$42,102
10	S2-O3-I1	\$110,000	\$110,000	\$220,000	\$362,708	26	\$142,708
11	S2-O3-I2	\$110,000	\$100,000	\$210,000	\$311,316	24	\$101,316
12	S2-O3-I3	\$110,000	\$205,000	\$315,000	\$272,646	20	-\$42,354
13	S3-O1-I1	\$95,000	\$150,000	\$245,000	\$349,636	26	\$104,636
14	S3-O1-I2	\$95,000	\$140,000	\$235,000	\$297,910	24	\$62,910
15	S3-O1-I3	\$95,000	\$255,000	\$350,000	\$258,804	20	-\$91,196
16	S3-O2-I1	\$95,000	\$125,000	\$220,000	\$329,136	24	\$109,136
17	S3-O2-I2	\$95,000	\$120,000	\$215,000	\$276,996	20	\$61,996
18	S3-O2-I3	\$95,000	\$250,000	\$345,000	\$251,808	18	-\$93,192
19	S3-O3-I1	\$95,000	\$105,000	\$200,000	\$308,804	22	\$108,804
20	S3-O3-I2	\$95,000	\$100,000	\$195,000	\$256,250	20	\$61,250
21	S3-O3-I3	\$95,000	\$225,000	\$320,000	\$230,522	16	-\$89,478

3.2 Reading the Cash vs Economic Gap

The "Cash vs Economic Gap" column shows the difference between total capital deployed (cash) and the full economic cost of the business. A positive gap means the economic cost exceeds cash invested — the difference is absorbed through deferred founder compensation, friendly-network discounts, and sweat equity. A negative gap means more cash is raised than the economic cost, providing working capital runway beyond what is strictly needed for build-out.

3.3 Capital Intensity by Scenario Family

Scenario Family	Avg Founder Capital	Avg Investor Capital	Avg Total Capital	Avg Economic Cost
S1 (self-funded, 3 paths)	\$191,667	\$0	\$191,667	\$362,492
S2 (investor at transition, 9 paths)	\$113,333	\$160,000	\$273,333	\$339,914
S3 (investor after Pilot 1, 9 paths)	\$95,000	\$163,333	\$258,333	\$284,429

4. Revenue Projections Summary

4.1 Revenue Streams and Pricing Corridors

Revenue Stream	Pilot 2 Price (50%)	Production Price (100%)	Margin
GP consultation	\$1.50-\$2.63	\$2.60-\$5.25	89-94%
Specialist consultation	\$2.63-\$5.25	\$5.25-\$10.50	94-96%
Pharmacy commission	4-6% of order	8-12% of order	94-97%
Diagnostics commission	5-7.5% of order	10-15% of order	94-97%
Video premium add-on	N/A	\$1.15-\$1.32/session	86-92%
Lab facilitation	N/A	\$1.15-\$1.32/order	94-97%
Rx coordination	N/A	\$0.62-\$0.70/order	91-95%
Premium subscription	\$2.31-\$2.63/mo	\$4.62-\$5.26/mo	78-88%
Facility platform fee	N/A	\$52.63/mo	~95%
Facility commission	N/A	12% of consult fees	~95%

4.2 ARPU by Phase (Moderate Assumptions)

Phase	ARPU/Active User/Month	Active User Definition
Pilot 2	\$2.65	35% of registered users, net of 7% monthly churn
Production	\$5.54	35% of registered users, net of 7% monthly churn

4.3 Revenue Trajectory by Representative Path

The following table shows monthly revenue at key milestones for representative paths within each scenario family. All figures in USD, moderate assumptions.

Combo	M6 Rev	M9 Rev	M12 Rev	M15 Rev	M18 Rev	M21 Rev	M24 Rev	Cum Rev M24
S1-L1	\$0	\$0	\$173	\$398	\$1,128	\$1,828	\$2,773	\$16,155
S1-L2	\$124	\$317	\$1,167	\$1,955	\$3,048	\$4,541	\$6,484	\$45,525
S1-L3	\$355	\$1,400	\$2,514	\$4,065	\$6,228	\$9,189	\$13,085	\$96,094
S2-O2-I2	\$148	\$511	\$1,910	\$3,578	\$6,152	\$9,855	\$14,920	\$94,812
S2-O3-I3	\$254	\$2,587	\$5,498	\$10,209	\$17,669	\$28,138	\$41,882	\$264,949
S3-O2-I2	\$310	\$1,372	\$2,680	\$5,095	\$8,834	\$14,462	\$22,378	\$132,724
S3-O3-I3	\$687	\$2,812	\$5,448	\$9,746	\$16,502	\$26,631	\$40,927	\$262,113

4.4 Revenue Comparison — All 21 Combinations at M24

#	Code	Monthly Rev M12	Monthly Rev M18	Monthly Rev M24	Cum Rev M24	CapEx M24
1	S1-L1	\$173	\$1,128	\$2,773	\$16,155	\$12,500
2	S1-L2	\$1,167	\$3,048	\$6,484	\$45,525	\$25,000
3	S1-L3	\$2,514	\$6,228	\$13,085	\$96,094	\$50,000
4	S2-O1-I1	\$275	\$2,054	\$5,543	\$33,908	\$54,500
5	S2-O1-I2	\$483	\$3,599	\$10,602	\$62,596	\$117,500
6	S2-O1-I3	\$603	\$5,432	\$17,669	\$107,579	\$255,000
7	S2-O2-I1	\$1,452	\$4,581	\$10,143	\$72,316	\$57,500
8	S2-O2-I2	\$1,910	\$6,152	\$14,920	\$94,812	\$126,500
9	S2-O2-I3	\$3,383	\$12,959	\$35,076	\$233,280	\$260,000
10	S2-O3-I1	\$2,835	\$7,271	\$14,679	\$105,399	\$85,000

11	S2-O3-I2	\$3,761	\$11,127	\$24,213	\$163,062	\$146,000
12	S2-O3-I3	\$5,498	\$17,669	\$41,882	\$264,949	\$274,000
13	S3-O1-I1	\$1,405	\$4,118	\$9,152	\$61,256	\$59,500
14	S3-O1-I2	\$2,319	\$7,658	\$19,632	\$116,967	\$126,000
15	S3-O1-I3	\$4,058	\$14,006	\$36,386	\$209,204	\$264,000
16	S3-O2-I1	\$1,818	\$5,271	\$11,729	\$76,655	\$62,500
17	S3-O2-I2	\$2,680	\$8,834	\$22,378	\$132,724	\$129,500
18	S3-O2-I3	\$5,039	\$17,669	\$44,494	\$266,975	\$270,000
19	S3-O3-I1	\$2,835	\$7,271	\$15,213	\$103,519	\$70,000
20	S3-O3-I2	\$4,541	\$14,006	\$32,720	\$199,342	\$138,000
21	S3-O3-I3	\$5,448	\$16,502	\$40,927	\$262,113	\$270,000

5. Break-Even Analysis

5.1 Definition of Break-Even Milestones

Three distinct break-even points are tracked for each combination:

- 1. First Revenue Month (FRM):** The month in which the platform collects its first non-zero revenue. This occurs at the start of Pilot 2 when 50% pricing is activated.
- 2. Monthly Cash-Flow Positive (MCFP):** The month in which monthly revenue first exceeds monthly burn rate (operating expenses for that month).
- 3. Cumulative Break-Even (CBE):** The month in which cumulative total revenue first exceeds cumulative total capital invested. This is the primary success metric defined in params.md Section 12.

5.2 Break-Even Milestones — S1 (Self-Funded)

Code	First Revenue Month	Monthly Burn at FRM	Monthly Rev > Monthly Burn	Cumulative Break-Even
S1-L1	M10	\$13,379/mo	M18 (\$6,307 rev vs \$13,379 burn) — not yet; projected M28-M30	M30-M36
S1-L2	M6	\$11,832/mo	M13 (\$6,643 rev vs \$13,711 burn) — not yet; projected M21-M23	M22-M24
S1-L3	M5	\$13,918/mo	M9 (\$1,400 rev vs \$15,191 burn) — not yet; projected M15-M17	M14-M16

Note: For S1 scenarios, the monthly burn rate during Pilot 2 and Production includes founder-office cost (\$5,200/mo), employed ops staff (\$2,482-\$6,542/mo), infrastructure (\$270-\$1,500/mo), and ongoing development. Revenue must surpass this entire monthly cost stack to achieve MCFP.

5.3 Break-Even Milestones — S2 (Investor During Transition)

Code	First Revenue Month	Investor Arrival	Monthly Rev > Monthly Burn	Cumulative Break-Even
S2-O1-I1	M10	M9	M26-M28	M28-M32
S2-O1-I2	M10	M9	M22-M24	M30-M34
S2-O1-I3	M10	M9	M18-M20	M30-M36
S2-O2-I1	M6	M7	M20-M22	M22-M26
S2-O2-I2	M6	M7	M18-M20	M26-M30
S2-O2-I3	M6	M7	M14-M16	M24-M28
S2-O3-I1	M5	M6	M15-M17	M20-M24
S2-O3-I2	M5	M6	M13-M15	M22-M24
S2-O3-I3	M5	M6	M10-M12	M22-M26

5.4 Break-Even Milestones — S3 (Investor After Pilot 1)

Code	First Revenue Month	Investor Arrival	Monthly Rev > Monthly Burn	Cumulative Break-Even
S3-O1-I1	M6	M3	M20-M22	M22-M26
S3-O1-I2	M5	M3	M16-M18	M24-M28
S3-O1-I3	M5	M3	M13-M15	M26-M30
S3-O2-I1	M5	M3	M18-M20	M18-M22
S3-O2-I2	M5	M3	M15-M17	M24-M26
S3-O2-I3	M5	M3	M11-M13	M24-M26
S3-O3-I1	M5	M3	M15-M17	M18-M22
S3-O3-I2	M5	M3	M12-M14	M22-M24
S3-O3-I3	M5	M3	M10-M12	M24-M25

5.5 Break-Even Summary Heat Map

The following table ranks all 21 combinations by cumulative break-even speed, from fastest to slowest:

Rank	Code	CBE Month Range	Total Capital at CBE	Category
1	S1-L3	M14-M16	\$30,000-\$34,000	Fastest: low capital, fast build
2	S1-L2	M16-M18	\$17,000-\$19,000	Fast: very low capital base
3	S3-O2-I1	M18-M22	\$52,000-\$62,500	Fast: modest investor, good pace
4	S3-O3-I1	M18-M22	\$54,000-\$67,500	Fast: full bridge, modest investor
5	S2-O3-I1	M20-M24	\$67,000-\$85,000	Good: strong self-fund, small investor
6	S1-L1	M22-M24	\$11,500-\$12,500	Good: tiny capital base, slow growth
7	S2-O2-I1	M22-M26	\$49,000-\$57,500	Good: moderate total capital
8	S2-O3-I2	M22-M24	\$122,000-\$146,000	Good: strong growth overcomes CapEx
9	S3-O1-I1	M22-M26	\$49,000-\$59,500	Moderate: constrained start
10	S3-O3-I2	M22-M24	\$120,000-\$138,000	Moderate: high growth
11	S2-O3-I3	M22-M26	\$242,000-\$274,000	Moderate: massive revenue overcomes massive CapEx
12	S3-O2-I2	M24-M26	\$114,500-\$129,500	Moderate: baseline investor path
13	S3-O3-I3	M24-M25	\$268,000-\$270,000	Moderate: maximum scenario, near M24 crossover
14	S3-O2-I3	M24-M26	\$253,000-\$270,000	Moderate: heavy investor
15	S3-O1-I2	M24-M28	\$112,000-\$126,000	Slower: constrained start, moderate investor
16	S2-O2-I2	M26-M30	\$119,000-\$126,500	Slower: baseline plan
17	S3-O1-I3	M26-M30	\$243,000-\$264,000	Slower: large CapEx from weak base
18	S2-O1-I1	M28-M32	\$50,500-\$54,500	Slow: constrained both sides
19	S2-O1-I2	M30-M34	\$108,000-\$117,500	Slow: late revenue start
20	S2-O1-I3	M30-M36	\$240,000-\$255,000	Slowest: large capital, late revenue
21	S1-L1 (extended)	M30-M36	\$15,500-\$18,500	Slowest by elapsed time

6. ROI by Scenario Family

6.1 S1 — Self-Funded (3 Paths)

In self-funded scenarios, founder ROI is the only relevant metric. The founder bears 100% of the economic cost and retains 60% equity (with 20% held by the founding team and 20% reserved but unissued).

ROI is measured at 12, 24, and 36 months after production launch begins (not from project start).

S1 Production Launch Timing

Code	Production Starts	M+12 Post-Prod	M+24 Post-Prod	M+36 Post-Prod
S1-L1	M22	M34	M46	M58
S1-L2	M16	M28	M40	M52
S1-L3	M12	M24	M36	M48

S1 ROI at Post-Production Milestones

Code	Founder Capital	Cum Rev at Prod+12	Cum Rev at Prod+24	Cum Rev at Prod+36	ROI at Prod+12	ROI at Prod+24	ROI at Prod+36
S1-L1	\$240,000	\$53,399	\$175,000-\$210,000	\$380,000-\$460,000	-78%	-27% to -12%	+58% to +92%
S1-L2	\$185,000	\$65,246	\$195,000-\$240,000	\$420,000-\$520,000	-65%	+5% to +30%	+127% to +181%
S1-L3	\$150,000	\$96,094	\$290,000-\$360,000	\$620,000-\$780,000	-36%	+93% to +140%	+313% to +420%

S1 Interpretation

- **S1-L1** is the slowest path. Its 34-month build timeline means that even 12 months post-production, cumulative revenue is still well below cumulative founder investment. Positive ROI does not arrive until approximately 30 months post-production (Month 52 from project start). However, the capital base is the lowest (\$240,000), so the downside risk is also the lowest.
- **S1-L2** is the balanced self-funded path. It reaches positive ROI around 18-24 months post-production. By Year 3 post-production (Month 40+), ROI reaches 127-181%, delivering a solid return for founder patience.
- **S1-L3** delivers the strongest percentage ROI among all S1 paths. The accelerated timeline means revenue compounds earlier and longer. By 36 months post-production, the founder has generated \$620K-\$780K in cumulative revenue against a \$150K investment — a 313-420% ROI.

6.2 S2 — Investor During Transition (9 Paths)

In S2 scenarios, the founder self-funds through Pre-Pilot and Pilot 1, then an investor arrives during the Pilot 1 to Pilot 2 transition. Founder and investor ROI are tracked separately.

Equity assumptions for S2 (per params.md Section 8.3):

- Founder retains 55-60% post-raise
- Founding team retains 18-20% post-raise
- Investor receives 15-20% from the investor pool

For modeling purposes, we use: Founder 58%, Founding Team 19%, Investor 18% (midpoint of ranges), with the remaining 5% reserved.

S2 Founder ROI (at 24mo and 36mo post-production launch)

Code	Founder Capital	Founder Equity	Cum Rev Prod+24	Founder Share Prod+24	Founder ROI Prod+24	Cum Rev Prod+36	Founder Share Prod+36	Founder ROI Prod+36
S2-O1-I1	\$120,000	58%	\$130,000-\$160,000	\$75,400-\$92,800	-37% to -23%	\$310,000-\$400,000	\$179,800-\$232,000	+50% to +93%
S2-O1-I2	\$120,000	58%	\$210,000-\$260,000	\$121,800-\$150,800	+2% to +26%	\$480,000-\$600,000	\$278,400-\$348,000	+132% to +190%
S2-O1-I3	\$120,000	58%	\$350,000-\$430,000	\$203,000-\$249,400	+69% to +108%	\$780,000-\$980,000	\$452,400-\$568,400	+277% to +374%
S2-O2-I1	\$110,000	58%	\$220,000-\$270,000	\$127,600-\$156,600	+16% to +42%	\$500,000-\$620,000	\$290,000-\$359,600	+164% to +227%
S2-O2-I2	\$110,000	58%	\$300,000-\$370,000	\$174,000-\$214,600	+58% to +95%	\$680,000-\$850,000	\$394,400-\$493,000	+259% to +348%
S2-O2-I3	\$110,000	58%	\$600,000-\$750,000	\$348,000-\$435,000	+216% to +295%	\$1,200,000-\$1,500,000	\$696,000-\$870,000	+533% to +691%
S2-O3-I1	\$110,000	58%	\$310,000-\$390,000	\$179,800-\$226,200	+63% to +106%	\$680,000-\$850,000	\$394,400-\$493,000	+259% to +348%
S2-O3-I2	\$110,000	58%	\$470,000-\$580,000	\$272,600-\$336,400	+148% to +206%	\$1,020,000-\$1,280,000	\$591,600-\$742,400	+438% to +575%
S2-O3-I3	\$110,000	58%	\$730,000-\$910,000	\$423,400-\$527,800	+285% to +380%	\$1,500,000-\$1,900,000	\$870,000-\$1,102,000	+691% to +902%

S2 Investor ROI (at 24mo and 36mo post-production launch)

Code	Investor Capital	Investor Equity	Investor Share Prod+24	Investor ROI Prod+24	Investor Share Prod+36	Investor ROI Prod+36
S2-O1-I1	\$140,000	18%	\$23,400-\$28,800	-83% to -79%	\$55,800-\$72,000	-60% to -49%
S2-O1-I2	\$145,000	18%	\$37,800-\$46,800	-74% to -68%	\$86,400-\$108,000	-40% to -26%
S2-O1-I3	\$245,000	18%	\$63,000-\$77,400	-74% to -68%	\$140,400-\$176,400	-43% to -28%
S2-O2-I1	\$135,000	18%	\$39,600-\$48,600	-71% to -64%	\$90,000-\$111,600	-33% to -17%
S2-O2-I2	\$120,000	18%	\$54,000-\$66,600	-55% to -45%	\$122,400-\$153,000	+2% to +28%
S2-O2-I3	\$240,000	18%	\$108,000-\$135,000	-55% to -44%	\$216,000-\$270,000	-10% to +13%
S2-O3-I1	\$110,000	18%	\$55,800-\$70,200	-49% to -36%	\$122,400-\$153,000	+11% to +39%
S2-O3-I2	\$100,000	18%	\$84,600-\$104,400	-15% to +4%	\$183,600-\$230,400	+84% to +130%
S2-O3-I3	\$205,000	18%	\$131,400-\$163,800	-36% to -20%	\$270,000-\$342,000	+32% to +67%

6.3 S3 — Investor After Pilot 1 (9 Paths)

In S3 scenarios, the founder self-funds only through Pre-Pilot and into Pilot 1 (0-3 months of self-funding). The investor arrives right after Pilot 1 completes, funding Pilot 2 and Production.

Equity assumptions for S3 (per params.md Section 8.3):

- Founder retains 52-58% post-raise
- Founding team retains 17-19% post-raise
- Investor receives 18-20% from the investor pool

For modeling purposes, we use: Founder 55%, Founding Team 18%, Investor 19% (midpoint reflecting slightly higher investor leverage in S3 due to earlier entry).

S3 Founder ROI (at 24mo and 36mo post-production launch)

Code	Founder Capital	Founder Equity	Cum Rev Prod+24	Founder Share Prod+24	Founder ROI Prod+24	Cum Rev Prod+36	Founder Share Prod+36	Founder ROI Prod+36
S3-O1-I1	\$95,000	55%	\$200,000-\$250,000	\$110,000-\$137,500	+16% to +45%	\$460,000-\$580,000	\$253,000-\$319,000	+166% to +236%
S3-O1-I2	\$95,000	55%	\$370,000-\$460,000	\$203,500-\$253,000	+114% to +166%	\$800,000-\$1,000,000	\$440,000-\$550,000	+363% to +479%

S3-O1-I3	\$95,000	55%	\$600,000-\$750,000	\$330,000-\$412,500	+247% to +334%	\$1,250,000-\$1,560,000	\$687,500-\$858,000	+624% to +803%
S3-O2-I1	\$95,000	55%	\$250,000-\$310,000	\$137,500-\$170,500	+45% to +79%	\$560,000-\$700,000	\$308,000-\$385,000	+224% to +305%
S3-O2-I2	\$95,000	55%	\$410,000-\$510,000	\$225,500-\$280,500	+137% to +195%	\$900,000-\$1,120,000	\$495,000-\$616,000	+421% to +548%
S3-O2-I3	\$95,000	55%	\$740,000-\$920,000	\$407,000-\$506,000	+328% to +433%	\$1,500,000-\$1,880,000	\$825,000-\$1,034,000	+768% to +988%
S3-O3-I1	\$95,000	55%	\$310,000-\$390,000	\$170,500-\$214,500	+79% to +126%	\$680,000-\$850,000	\$374,000-\$467,500	+294% to +392%
S3-O3-I2	\$95,000	55%	\$580,000-\$720,000	\$319,000-\$396,000	+236% to +317%	\$1,250,000-\$1,560,000	\$687,500-\$858,000	+624% to +803%
S3-O3-I3	\$95,000	55%	\$750,000-\$930,000	\$412,500-\$511,500	+334% to +438%	\$1,550,000-\$1,940,000	\$852,500-\$1,067,000	+797% to +1023%

S3 Investor ROI (at 24mo and 36mo post-production launch)

Code	Investor Capital	Investor Equity	Investor Share Prod+24	Investor ROI Prod+24	Investor Share Prod+36	Investor ROI Prod+36
S3-O1-I1	\$150,000	19%	\$38,000-\$47,500	-75% to -68%	\$87,400-\$110,200	-42% to -27%
S3-O1-I2	\$140,000	19%	\$70,300-\$87,400	-50% to -38%	\$152,000-\$190,000	+9% to +36%
S3-O1-I3	\$255,000	19%	\$114,000-\$142,500	-55% to -44%	\$237,500-\$296,400	-7% to +16%
S3-O2-I1	\$125,000	19%	\$47,500-\$58,900	-62% to -53%	\$106,400-\$133,000	-15% to +6%
S3-O2-I2	\$120,000	19%	\$77,900-\$96,900	-35% to -19%	\$171,000-\$212,800	+43% to +77%
S3-O2-I3	\$250,000	19%	\$140,600-\$174,800	-44% to -30%	\$285,000-\$357,200	+14% to +43%
S3-O3-I1	\$105,000	19%	\$58,900-\$74,100	-44% to -29%	\$129,200-\$161,500	+23% to +54%
S3-O3-I2	\$100,000	19%	\$110,200-\$136,800	+10% to +37%	\$237,500-\$296,400	+138% to +196%
S3-O3-I3	\$225,000	19%	\$142,500-\$176,700	-37% to -21%	\$294,500-\$368,600	+31% to +64%

7. Payback Period Analysis

7.1 Definition

Payback period is the number of months from the time capital is deployed until the cumulative revenue attributable to that stakeholder (based on their equity share) equals their capital contribution.

For founders, the clock starts at M0 (project kickoff). For investors, the clock starts at the month they deploy capital.

7.2 Founder Payback Period — All 21 Combinations

#	Code	Founder Capital	Founder Equity Share	Months to Payback	Calendar Month
1	S1-L1	\$240,000	60%	46-52	M46-M52
2	S1-L2	\$185,000	60%	34-40	M34-M40
3	S1-L3	\$150,000	60%	24-30	M24-M30
4	S2-O1-I1	\$120,000	58%	36-42	M36-M42
5	S2-O1-I2	\$120,000	58%	30-36	M30-M36
6	S2-O1-I3	\$120,000	58%	24-28	M24-M28
7	S2-O2-I1	\$110,000	58%	28-34	M28-M34
8	S2-O2-I2	\$110,000	58%	24-30	M24-M30
9	S2-O2-I3	\$110,000	58%	18-22	M18-M22
10	S2-O3-I1	\$110,000	58%	22-28	M22-M28
11	S2-O3-I2	\$110,000	58%	20-24	M20-M24
12	S2-O3-I3	\$110,000	58%	16-20	M16-M20
13	S3-O1-I1	\$95,000	55%	28-34	M28-M34
14	S3-O1-I2	\$95,000	55%	22-26	M22-M26
15	S3-O1-I3	\$95,000	55%	18-22	M18-M22
16	S3-O2-I1	\$95,000	55%	24-28	M24-M28
17	S3-O2-I2	\$95,000	55%	20-24	M20-M24

18	S3-O2-I3	\$95,000	55%	16-18	M16-M18
19	S3-O3-I1	\$95,000	55%	22-26	M22-M26
20	S3-O3-I2	\$95,000	55%	18-22	M18-M22
21	S3-O3-I3	\$95,000	55%	14-18	M14-M18

7.3 Investor Payback Period — S2 and S3 Combinations

#	Code	Investor Capital	Investor Equity	Investor Entry	Months to Payback (from entry)	Calendar Month
4	S2-O1-I1	\$140,000	18%	M9	40-48+	M49-M57+
5	S2-O1-I2	\$145,000	18%	M9	36-44	M45-M53
6	S2-O1-I3	\$245,000	18%	M9	36-44	M45-M53
7	S2-O2-I1	\$135,000	18%	M7	34-42	M41-M49
8	S2-O2-I2	\$120,000	18%	M7	28-36	M35-M43
9	S2-O2-I3	\$240,000	18%	M7	30-38	M37-M45
10	S2-O3-I1	\$110,000	18%	M6	26-34	M32-M40
11	S2-O3-I2	\$100,000	18%	M6	22-28	M28-M34
12	S2-O3-I3	\$205,000	18%	M6	26-32	M32-M38
13	S3-O1-I1	\$150,000	19%	M3	38-46	M41-M49
14	S3-O1-I2	\$140,000	19%	M3	30-36	M33-M39
15	S3-O1-I3	\$255,000	19%	M3	32-40	M35-M43
16	S3-O2-I1	\$125,000	19%	M3	30-36	M33-M39
17	S3-O2-I2	\$120,000	19%	M3	26-32	M29-M35
18	S3-O2-I3	\$250,000	19%	M3	28-34	M31-M37
19	S3-O3-I1	\$105,000	19%	M3	24-30	M27-M33
20	S3-O3-I2	\$100,000	19%	M3	20-26	M23-M29
21	S3-O3-I3	\$225,000	19%	M3	24-30	M27-M33

7.4 Payback Period Key Findings

Finding	Detail
Fastest founder payback	S3-O3-I3: 14-18 months (founder invests \$95K, gets a fully funded operation that generates revenue fast)
Fastest investor payback	S3-O3-I2: 20-26 months from investor entry (investor puts in \$100K for 19% equity in a well-funded, fast-growing operation)
Slowest founder payback	S1-L1: 46-52 months (long self-funded grind with slow revenue ramp)
Slowest investor payback	S2-O1-I1: 40-48+ months (investor enters late after slow build, revenue growth is modest)
Best investor capital efficiency	S2-O3-I2 and S3-O3-I2: \$100K invested, payback in 20-28 months — the ideal "right-sized" investor
Worst investor capital efficiency	S2-O1-I3: \$245K invested into a slow-build operation, payback 36-44 months

8. Sensitivity Analysis

8.1 Scenario Definitions

ROI projections are stress-tested under three revenue assumption sets:

Parameter	Pessimistic (50% Rev)	Baseline (100% Rev)	Optimistic (150% Rev)
Revenue multiplier	0.5x of moderate projection	1.0x (moderate)	1.5x of moderate projection
Monthly user growth	7.5%	15%	22.5%
Active user rate	17.5% of registered	35% of registered	52.5% of registered
ARPU (Production)	\$2.77	\$5.54	\$8.31
Monthly churn	10%	7%	4%

Costs are held constant across all three scenarios — the investment is already committed regardless of revenue performance.

8.2 Sensitivity Matrix — S1 (Self-Funded) at M36 Post-Production

Code	Pessimistic Cum Rev	Pessimistic ROI	Baseline Cum Rev	Baseline ROI	Optimistic Cum Rev	Optimistic ROI
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S1-L1	\$95,000-\$115,000	-60% to -52%	\$190,000-\$230,000	-21% to -4%	\$285,000-\$345,000	+19% to +44%
S1-L2	\$210,000-\$260,000	+14% to +41%	\$420,000-\$520,000	+127% to +181%	\$630,000-\$780,000	+241% to +322%
S1-L3	\$310,000-\$390,000	+107% to +160%	\$620,000-\$780,000	+313% to +420%	\$930,000-\$1,170,000	+520% to +680%

8.3 Sensitivity Matrix — S2 Representative Paths at M36 Post-Production (Combined ROI)

Code	Pessimistic ROI	Baseline ROI	Optimistic ROI
S2-O1-I1	-65% to -55%	-10% to +10%	+40% to +70%
S2-O2-I2	-30% to -15%	+50% to +80%	+140% to +190%
S2-O3-I2	+10% to +30%	+110% to +150%	+220% to +290%
S2-O3-I3	+30% to +50%	+170% to +220%	+330% to +420%

8.4 Sensitivity Matrix — S3 Representative Paths at M36 Post-Production (Combined ROI)

Code	Pessimistic ROI	Baseline ROI	Optimistic ROI
S3-O1-I1	-55% to -40%	+5% to +25%	+60% to +100%
S3-O2-I2	-15% to +5%	+70% to +100%	+170% to +230%
S3-O3-I2	+20% to +40%	+140% to +190%	+280% to +370%
S3-O3-I3	+25% to +45%	+160% to +210%	+310% to +410%

8.5 Sensitivity — Impact of Key Variables on Baseline (S2-O2-I2)

The following table isolates the impact of changing a single variable while holding all others at baseline:

Variable Changed	Direction	Impact on M36 Cumulative Revenue	Impact on Combined ROI
ARPU +25%	Positive	+\$85,000 to +\$120,000	+25 to +35 pp
ARPU -25%	Negative	-\$85,000 to -\$120,000	-25 to -35 pp
Monthly growth rate +5pp	Positive	+\$150,000 to +\$200,000	+45 to +60 pp
Monthly growth rate -5pp	Negative	-\$100,000 to -\$140,000	-30 to -45 pp
Churn rate +3pp	Negative	-\$70,000 to -\$100,000	-20 to -30 pp
Churn rate -3pp	Positive	+\$90,000 to +\$130,000	+25 to +40 pp
Production launch delayed 3mo	Negative	-\$50,000 to -\$80,000	-15 to -25 pp
Production launch accelerated 3mo	Positive	+\$60,000 to +\$90,000	+18 to +28 pp
FX depreciation +10%	Negative	-\$30,000 to -\$50,000	-10 to -15 pp
Facility onboarding +50%	Positive	+\$40,000 to +\$60,000	+12 to +18 pp

8.6 Breakeven Probability by Scenario (36-month window)

Based on the sensitivity ranges, the estimated probability of achieving cumulative breakeven within 36 months of project start:

Code	Pessimistic	Baseline	Optimistic
S1-L1	15%	55%	85%
S1-L2	60%	90%	98%
S1-L3	85%	98%	>99%
S2-O2-I2	35%	75%	95%
S2-O3-I3	60%	90%	98%
S3-O2-I2	45%	80%	95%
S3-O3-I2	65%	92%	98%
S3-O3-I3	55%	88%	97%

9. Non-Financial ROI

9.1 Strategic Value Assets

Beyond direct revenue returns, Health Hub generates strategic value that is difficult to quantify in percentage terms but materially affects the company's long-term positioning and exit potential.

Market Position Value

Asset	Description	Strategic Significance
First-mover advantage in Ethiopia digital health	No established digital health marketplace exists in Ethiopia at scale	Regulatory relationships, clinic partnerships, and patient trust are defensible moats that take years to replicate
Multi-stakeholder platform architecture	Serves patients, GPs, specialists, pharmacies, diagnostics, and admins simultaneously	Network effects create exponential value — each new stakeholder type increases value for all others
Dual-country operating capability	Ethiopia + Kenya provides geographic diversification and proof of cross-border scalability	Significantly increases attractiveness for Series A investors who require evidence of market expansion
Regulatory pathway documentation	Compliance and regulatory documentation developed during Pre-Pilot and Pilot phases	Reduces time-to-market for subsequent country entries from 12-18 months to 6-8 months

Data Asset Value

Data Type	When Accumulated	Strategic Use
Patient health interaction patterns	Pilot 1+	AI triage development, predictive health, personalized care pathways
Facility utilization and capacity data	Pilot 2+	Supply-side optimization, dynamic pricing, capacity planning
Treatment outcome tracking	Production+	Evidence-based care protocols, quality scoring, outcomes-based contracting
Regional disease prevalence patterns	Production+	Public health partnerships, insurance product design, pharmaceutical partnerships
Payment and affordability data	Pilot 2+	Pricing optimization, micro-insurance product design, credit scoring

The anonymized data asset alone could generate \$50,000-\$200,000/year in licensing revenue by Year 3-4 (listed as an extension revenue stream in 05-revenue-modelling). More importantly, the data creates a defensible competitive moat that increases with every transaction processed.

Network Effects

Network Effect Type	Mechanism	Value Creation
Cross-side (patient-to-provider)	More patients attract more providers; more providers attract more patients	Each additional clinic increases available appointment slots, reducing wait times and improving patient retention
Same-side (provider-to-provider)	GP referrals to specialists create a closed loop within the platform	Referral network density increases switching costs for providers and keeps revenue within the ecosystem
Data network effects	More transactions improve AI triage accuracy, which improves patient outcomes, which drives more transactions	Self-reinforcing cycle that creates exponentially increasing value with linear user growth
Geographic network effects	Success in Addis Ababa creates a replicable playbook for Nairobi and beyond	Operating costs per new city decrease by 30-50% after the first market is established

Brand Equity

Brand Dimension	How Built	ROI Implication
Trust in digital healthcare	Consistent service quality during Pilot phases; no data breaches; reliable doctor availability	Reduces customer acquisition cost by 20-40% in Year 2-3 as word-of-mouth replaces paid acquisition
Healthcare authority	Doctor training protocols, clinical oversight, quality scoring	Enables premium pricing and attracts institutional partnerships (hospitals, insurers, NGOs)
Technology credibility	SSR web platform, native Android app, video consultations, real-time notifications	Differentiates from SMS-based or basic-web competitors; justifies platform fees to facilities

9.2 Exit Valuation Implications

While this document does not model exit-based ROI, the following benchmarks provide context for what Health Hub might be worth at various stages:

Stage	Comparable Valuation Multiple	Implied Valuation Range	Basis
End of Pilot 2 (1K-10K users)	5-10x trailing annual revenue	\$150K-\$1.5M	Pre-seed/seed stage digital health in Africa
Production Year 1 (10K-50K users)	8-15x trailing annual revenue	\$1M-\$7.5M	Seed/Series A for emerging market health tech
Production Year 2 (50K-200K users)	12-20x trailing annual revenue	\$5M-\$30M	Series A/B comparable (mPharma, Helium Health)
Production Year 3+ (200K+ users)	15-25x trailing annual revenue	\$15M-\$100M+	Growth-stage emerging market health platform

For an investor entering at S3 with \$100K-\$250K for 18-20% equity, a \$5M-\$10M valuation at Series A (Year 2-3) would represent a 2x-5x multiple on invested capital — competitive with early-stage venture returns. A \$30M+ valuation at Series B would represent a 10x-25x multiple.

9.3 Social Impact ROI

Impact Metric	Target by Production Year 2	Measurement
Patients receiving first-ever telemedicine consultation	5,000-20,000	Platform registration + first consultation completion
Reduction in patient travel time for routine care	60-80% for connected patients	Self-reported survey vs baseline

Specialist access in underserved areas	500-2,000 patients receiving specialist consults who otherwise would not	Referral completion from rural/peri-urban locations
Pharmacy accuracy improvement	15-25% reduction in prescription errors	Error tracking before/after platform-assisted dispensing
Health data digitization	10,000-50,000 patient records digitized	Cumulative patient profiles with visit history

Social impact metrics are increasingly important for impact-focused investors, DFI (Development Finance Institution) grant eligibility, and partnerships with organizations like WHO, USAID, and the Gates Foundation.

10. Key Takeaways

10.1 Best ROI Paths by Stakeholder

Stakeholder	Recommended Path	Why
Founder (maximizing % ROI)	S3-O3-I3	Lowest founder capital (\$95K), maximum investor leverage, fastest timeline (16 months), 797-1023% founder ROI at 36mo post-production
Founder (balancing ROI and control)	S3-O2-I2	\$95K founder capital, \$120K investor, 20-month timeline, 421-548% founder ROI at 36mo post-production, founder retains 55% equity
Founder (maximizing control)	S1-L3	No dilution, 60% retained, 20-month timeline, 313-420% ROI at 36mo post-production, but requires \$150K self-funding
Investor (maximizing % return)	S3-O3-I2	\$100K for 19% equity, 20-26 month payback, 138-196% ROI at 36mo post-production, well-funded operation reduces risk
Investor (balancing return and risk)	S3-O2-I2	\$120K for 19% equity, 26-32 month payback, 43-77% ROI at 36mo post-production, moderate but reliable growth
Investor (maximizing absolute return)	S3-O3-I3 or S2-O3-I3	Largest capital deployed (\$205K-\$225K), largest absolute dollar returns, but lower percentage ROI and longer payback

10.2 Paths to Avoid

Path	Reason to Avoid
S2-O1-I3	\$245K investor capital enters after a slow L1 self-funded phase. Capital sits idle for months. Investor ROI is negative at 36mo post-production under pessimistic assumptions.
S2-O1-I1	Both founder and investor are constrained. Slowest investor payback (40-48+ months). Insufficient capital to drive meaningful growth.
S1-L1	34-month timeline to production. Founder payback takes 46-52 months. High risk of founder fatigue. Only 55% probability of breakeven within 36 months under pessimistic assumptions.
Any O1-I3 combination	The mismatch between a bootstrapped self-funded phase and a large investor injection creates inefficiency — the strong investor capital cannot compensate for the weak product foundation built during L1.

10.3 Recommended Path

S3-O2-I2 (Steady Bridge + Ideal Investor) is the recommended path for the following reasons:

Criterion	S3-O2-I2 Score
Total capital required	\$215,000 (\$95K founder + \$120K investor) — achievable
Timeline to production	20 months — ambitious but realistic
Founder ROI at 36mo post-prod	421-548% — strong returns
Investor ROI at 36mo post-prod	43-77% — competitive with early-stage venture
Investor payback period	26-32 months from entry — reasonable for pre-seed
Breakeven probability (36mo)	80% baseline, 95% optimistic
Founder equity retained	55% — clear majority control
Monthly revenue at M24	\$22,378 — demonstrates meaningful traction
Risk profile	Moderate — investor enters early enough to accelerate growth, but total capital is not so large that downside losses are catastrophic

If the founder's risk tolerance is higher and a larger investor is available, **S3-O3-I2** offers faster founder payback (18-22 months) and higher investor ROI (138-196% at 36mo post-production) while still keeping total capital at a manageable \$195,000.

10.4 ROI Drivers to Monitor

The following metrics most directly impact realized ROI and should be tracked monthly once revenue collection begins:

Metric	Target	Impact if Missed
Monthly active user growth rate	15%+	Every 1pp below target delays breakeven by ~1 month
Active user percentage	35%+ of registered	Below 25% indicates retention problem; halves ARPU

Monthly churn rate	Below 7%	Every 1pp above target reduces LTV by ~15%
ARPU (Production)	\$5.50+	Below \$3.50 shifts all scenarios 6-12 months right
Facility onboarding rate	2-3 per month at Production	Below 1/month caps B2B revenue and slows network effects
GP consultation volume	800+ per month at Production	Below 500 indicates demand or availability problem
Specialist referral conversion	40%+ of GP-referred patients see a specialist	Below 25% breaks the referral flywheel

10.5 Final Perspective

Health Hub's ROI profile is characteristic of a pre-revenue marketplace startup in an emerging market. The numbers show that:

1. **Capital efficiency is inversely correlated with absolute returns.** The cheapest paths (S1-L1, S1-L2) have the highest ROI percentages but generate the least absolute revenue and take the longest to reach meaningful scale.
2. **Investor capital accelerates revenue but creates a larger hurdle.** Every dollar of investor capital must be earned back through the investor's equity share, which is only 18-19%. This means \$1 of investor capital requires \$5.26-\$5.56 of total platform revenue to repay.
3. **The optimal strategy is right-sized investment.** Neither minimum nor maximum investment produces the best risk-adjusted returns. The I2 (Ideal) investor level consistently outperforms I1 (too constrained) and I3 (too much capital for the current stage) on ROI efficiency metrics.
4. **Time is the most expensive input.** The longest timelines (S1-L1 at 34 months, S2-O1-I1 at 30 months) accumulate the highest economic costs because founder-office time, ops staff, and infrastructure compound monthly. Faster paths are not just better for morale — they are structurally cheaper.
5. **Non-financial returns may exceed financial returns for early investors.** An investor who enters at \$100K-\$120K for 18-19% equity in a company that achieves \$5M-\$10M Series A valuation within 3 years would realize a 10-20x return — far exceeding the organic revenue-based ROI modeled in this document.

End of 07-roi-analysis.md. All figures are forward projections based on assumptions defined in [params.md](#). Actual results will vary. This document should be read alongside [05-revenue-modelling](#) and [03-cost-breakdown](#) for full context.